**Which of the statements given above is/are correct?**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**Ans. (c):** Both given statements are correct. The duration of the monsoon decreases from southern India to northern India. The amount of annual rainfall in the northern plains of India decreases from east to west. Since the rainfall in the Indian sub-continent is caused by moisture- bearing south-west monsoon winds, the duration of the monsoon is highest in nearby coastal regions and decreases onwards once the wind starts moving towards the inland regions. This happens due to the fast loosening of moisture in these winds. The same could be associated with the annual rainfall distribution in northern plains from east to west.

**18. A particular State in India has the following characteristics:**

1. It is located on the same latitude which passes through northern Rajasthan.
2. It has over 80% of its area under forest cover.
3. Over 12% of forest cover constitutes Protected Area Network in this State.

**Which one among the following States has all the above characteristics?**

(a) Arunachal Pradesh (b) Assam
(c) Himachal Pradesh (d) Uttarakhand

**Ans. (a):** Arunachal Pradesh is located on the same latitude which passes through northern Rajasthan. More than 12% of the forested area is in the form of a protected area network of this state and more than 80% area of this state comes under forest cover.

**19. With reference to the wetlands of India, consider the following statements:**

1. The country's total geographical area under the category of wetlands is recorded more in Gujarat as compared to other States.
2. In India, the total geographical area of coastal wetlands is larger than that of inland wetlands.

**Which of the statements given above is /are correct?**

(a) 1only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**Ans. (c):** The country's total geographical area under the wetlands in recorded highest in Gujarat as compared to other states. In India, the total geographical area of coastal wetlands in lesser than that of inland wetlands.

**20. When you travel in Himalayas, you will see the following ?**

1. Deep gorges
2. U-turn river courses
3. Parallel mountain ranges
4. Steep gradients causing land-sliding

**Which of the above can be said to be the evidences for Himalayas being young fold mountains?**

(a) 1 and 2 only (b) 1, 2 and 4 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

**Ans. (d):** Young fold mountains are rising under the influence of the earth's tectonic forces. They have a variety of rock structures, deep gorges and high pyramidal peaks. In the high Himalayas, the rivers have steep gradients, which result from the differential uplift of the high Himalayas. It suggested a long and narrow area of High Himalayas being uplifted during quaternary. Then, the Himalayas are called young fold mountains because parallel ranges are found here. Finally, all statements are correct. The correct answer is option (d).

## • Monsoon, Forest, Soils, Irrigation Projects, Agriculture

**21. Consider the following crops of India:**

1. Cowpea
2. Green Gram
3. Pigeon pea

**Which of the above is /are used as pulse, fodder and green manure?**

(a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

**Ans. (d):** Green Manure Crops or legumes like cowpea, horse gram and green gram improves precipitation- use efficiency during off- season. Cow - pea is Lobia. Lobia as green manure puts in nitrogen, builds organic matter, smothers weeds and controls nematodes. Pigeon pea (Tur/Arhar) is also extensively used in green manure. Pulses and Fodder are all of them. So the correct answer is option (d).

**22. Consider the following crops of India :**

1. Groundnut
2. Sesamum
3. Pearl millet

**Which of the above is /are predominantly rain-fed crop/crops ?**

(a) 1 and 2 only (b) 2 and 3
(c) 3 only (d) 1, 2 and 3

**Ans. (d) :** Groundnut, sesamum and pearl millet are cultivated in dry agricultural areas. These crops are mainly rainfed crops.

## • Minerals, Power Resources & Industry and Trade

**23. To meet its rapidly growing energy demand, some opine that India should pursue research and development on thorium as the future fuel of nuclear energy. In this context, what advantage does thorium hold over uranium ?**

1. Thorium is far more abundant in nature than uranium.
2. On the basis of per unit mass of mined mineral, thorium can generate more energy compared to natural uranium.
3. Thorium produces less harmful waste compared to uranium.

**Which of the statements given above is/are correct?**

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3



**Ans. (d):** Thorium could be used to fuel nuclear reactors, just like Uranium. Thorium is three times more abundant in nature than uranium, is not fissile on its own (which means reactions can be stopped when necessary), produces waste products that are less radioactive, and generates more energy per ton. All statement are correct.

**24. Despite having large reserves of coal, why does India import millions of tonnes of coal ?**

1. It is the policy of India to save its own coal reserves for future, and import it from other countries for the present use.
2. Most of the power plants in India are coal –based and they are not able to get sufficient supplies of coal from within the country.
3. Steel companies need large quantity of coking coal which had to be imported.

**Which of the statements given above is /are correct**

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

**Ans. (b) :** Indian imports several million tonnes of coal. Despite being a large reserve of coal, as steel companies require large amounts of coke coal and most of India's power plants are based on coal and have a substantial amount of coal that could not be supplied internally.

# WORLD GEOGRAPHY

## • Universe and Solar System

**25. Normally, the temperature decreases with the increase in height from the Earth's surface, because.**

1. The atmosphere can be heated upwards only from the Earth's surface.
2. There is more moisture in the upper atmosphere.
3. The air is less dense in the upper atmosphere.

**Select the correct answer using the codes given below:**

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

**Ans. (c):** Generally, the temperature decreases as the altitude increases from the earth's surface because the atmosphere can only warm upward from the Earth's surface and the air in the upper atmosphere is less dense. The atmosphere is mainly composed of gases, water vapour and dust. The amount of water vapour in the atmosphere is between 0 to 5%.

## • Grasslands/Lithosphere

**26. Which one of the following is the characteristic climate of the Tropical Savannah Region ?**

(a) Rainfall throughout the year
(b) Rainfall in winter only
(c) An extremely short dry season
(d) A definite dry and wet season

**Ans. (d):** Tropical Savannah has a definite dry and wet season. The wet summer season lasts 6-8 months and during these days, there is plenty of rainfall. Winter lasts for 4-6 months and there might be no rain in winter. This winter is the dry season in which there are frequent forest fires.

**27. What is the difference between the antelopes Oryx and Chiru?**

(a) Oryx is adapted to live in hot and arid areas whereas Chiru is adapted to live in steppes and semi-desert areas of cold height mountains.
(b) Oryx is poached for its antlers whereas Chiru is poached for its musk.
(c) Oryx exists in western India only whereas Chiru exists in north-east India only.
(d) None of the statements (a), (b) and (c) given above is correct.

**Ans. (a):** The Oryx is well-known for living in hot and dry areas, while Chiru is adapted to live in cold, high mountain meadows and semi-desert areas.

**28. Consider the following factors :**

1. Rotation of the Earth
2. Air pressure and wind
3. Density of ocean water
4. Revolution of the Earth

**Which of the above factors influence the ocean currents?**

(a) 1 and 2 only (b) 1, 2 and 3
(c) 1 and 4 (d) 2, 3 and 4

**Ans. (b):** Earth's rotation, air pressure, wind and the density of ocean water affect ocean currents. It is an important physical property of ocean water. It is determined by the amount of sunset, heat balance, density of ocean water, salinity, evaporation and condensation, local seasonal conditions, etc.

## • Hydrosphere

**29. What would happen if phytoplankton of an ocean is completely destroyed for some reason?**

1. The ocean as a carbon sink would be adversely affected.
2. The food chains in the ocean would be adversely affected.
3. The density of ocean water would drastically decrease.

**Select the correct answer using the codes given below :**

(a) 1 and 2 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

**Ans. (a) :** If the phytoplankton of an ocean is destroyed due to some reason, it will have the effect that the ocean will be adversely affected in the form of carbon sinks and the food chain of the ocean will be adversely affected.

**30. The acidification of oceans is increasing . Why is this phenomenon a cause of concern ?**

1. The growth and survival of calcareous phytoplankton will be adversely affected.
2. The growth and survival of coral reefs will be adversely affected.
3. The survival of some animals that have phytoplankton larvae will be adversely affected.
4. The cloud seeding and formation of clouds will be adversely affected.